Lemma 2.46 (Curve selection)**.** *Suppose that* $\mathbb{S}$ *is* VF*-generated. Let* $A \subseteq \mathrm{VF}^n$ *be definable and let* b *be a point in its closure in the coordinatewise projective line* $(\mathbb{P}^1(\mathrm{VF}))^n$*, with the valuation topology. Then there is a* b*-definable continuous finitary function* $g : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{P}(A)$ *whose values are nonempty and such that*

$$\lim_{\mathcal{M}\setminus\{0\} \longrightarrow 0} g = \{b\}.$$

Resolved referee R165 [Clarification; new] – Definability and nonempty values in curve selection.
Original source line 1081. State that A is definable. In the proof require $g(c) \neq \varnothing$ for every c (or at least on a punctured neighborhood of 0): "finitary" in the earlier definition allows empty values, and $g(c) \subseteq \mathfrak{o}(b, \mathrm{val}(c))$ alone is also satisfied by the identically empty correspondence. The algebraic-closure construction, once its base hypothesis is corrected below, supplies finite nonempty values and hence the stronger conclusion $\lim g = \{b\}$.

Author resolution. The author approves explicitly taking $A \subseteq \mathrm{VF}^n$ definable and requiring finite nonempty values at every input. The construction gives the stronger singleton limit $\{b\}$. Projective charts handle infinite coordinates without changing the affine target set of the curve.

Proof. Use a coordinatewise projective chart around b to reduce to the case $b \in \mathrm{VF}^n$: at an infinite coordinate use inversion and restrict A to the chart. These changes are definable over b and preserve convergence and continuity locally.

Naming the finite tuple b and a nonzero $c \in \mathcal{M}$ preserves VF-generation and the standing nontriviality of the ground valuation. Hence $\mathrm{acl}(\mathbb{S}, b, c)$ is a model of $\mathrm{ACVF}(0,0)$. Since $b \in \overline{A}$, the definable set $A \cap \mathfrak{o}(b, \mathrm{val}(c))$ is nonempty, so elementarity gives a point of that intersection algebraic over $(\mathbb{S}, b, c)$. Compactness supplies a b-definable finitary function g_0 on $\mathcal{M} \setminus \{0\}$ such that

$$\varnothing \neq g_0(c) \subseteq A \cap \mathfrak{o}(b, \mathrm{val}(c))$$

for every c. Thus g_0 converges to $\{b\}$, and nonempty values make this limit minimal.

By Lemma 2.44, with its one-dimensional source, g_0 is continuous outside a finite set. Choose a b-definable clopen disc $D \subseteq \mathcal{M}$ around 0 such that g_0 is continuous on $D \setminus \{0\}$. Also $\mathrm{acl}(\mathbb{S}, b)$ is a model, so A contains a finite nonempty b-definable set E. Define

$$g(c) = \begin{cases} g_0(c), & c \in D \setminus \{0\}, \\ E, & c \in \mathcal{M} \setminus D. \end{cases}$$

The two pieces are relatively clopen in $\mathcal{M} \setminus \{0\}$, so g is continuous, finite-valued and nonempty everywhere. Its limit at 0 is still $\{b\}$. Finally transport the construction back through the projective chart. This avoids assuming that a definable value-group radius is the valuation of a definable scalar. □

Resolved referee R040 [Error] – Curve selection requires a restriction on the base.
Original source line 1088. The assertion is false for the arbitrary $\mathbb{S}$ allowed so far. Let K be a nontrivially valued algebraically closed base field, add an RV parameter t with $\mathrm{vrv}(t) = 0$ transcendental over its residue field, and put $A = \{(x, y) : y \neq 0, \mathrm{rv}(x/y) = t\}$. Then $(0,0) \in \overline{A}$. A nonempty finite $\mathbb{S}$-definable choice in A at an input $c \in K \cap (\mathcal{M} \setminus \{0\})$ would have coordinates algebraic over K, t, c, hence in K by Lemma 2.13; their ratio cannot have leading term t. Even though finitary values may be empty, $b \in \lim g$ forces their support to accumulate at 0; by C-minimality that support contains a punctured neighborhood, whose defining radius lies in $\Gamma(K)$ since adjoining the residue parameter t does not enlarge definable values. It therefore contains such an input $c \in K$. Thus the claimed curve does not exist over that base. Impose VF-generation before this lemma (as already suggested in the human report), or allow